

VALEN LI

267-752-9395 | li.valen.008@gmail.com | [linkedin.com/valen-li](https://www.linkedin.com/in/valen-li) | github.com/li-valen

EDUCATION

AI + Finance @ UPenn's Jerome Fischer M&T Program (Class of 2030) Philadelphia, PA
Julia R. Masterman High School Philadelphia, PA
*GPA: 4.52/4.0 — Dual Enrollment: **Discrete Mathematics**, STAT 0001* Sep 2018 – June 2026
Robotics Team (Co-Founder & Lead Programmer) | Chess Team (Co-Captain) | AI Club (President)

EXPERIENCE

Wharton's Stevens Center for Innovation in Finance Philadelphia, PA
Full-Stack App Developer Jun 2024 – Jun 2025

- Enhanced key features of a student loan application to accurately project college costs, using PostgreSQL for data management and Spring Boot for API development
- Created a responsive [website](#) for the Finiverse project, and created web features for a data visualization platform.

Carnegie Mellon University Remote
Cyber-Security Researcher Feb 2025 – August 2025

- Worked with professor Hanan Hibshi on: The Practical Challenge of Writing Secure Code
- Researched using Self-Adapting Language Models (SEAL) for improving email phishing filters.
- Created an AI-powered filter that can autonomously learn from new phishing tactics.

University of Washington Remote
Machine Learning Researcher June 2024 – Aug 2024

- Trained ML models such as XGBoost, LightGBM, CatBoost, Random Forest, K-Means Clustering, and GNNs on an emerging alternative (SMBJ) to traditional PCR techniques used in genetic sequencing and identification.
- Built an AI system to classify COVID-19 variants from Single Molecule Break Junction (SMBJ) conductance signatures; achieved up to 99.97% accuracy (CatBoost).
- Developed a pipeline: filtering, R²-based QC, conductance conversion, and 1D/2D histogram features.

PROJECTS

LearnAI | *Computer Vision, Javascript, GSAP* Feb 2024 – May 2024

- Awarded **\$2,500** at a Hack-a-thon by creating a website redefining AI education with
- Built from scratch utilizing GSAP and Lenis for animations and smooth-scrolling also mimicking features from Apple's website.

Micro Pantry Mobile App | *Computer Vision, React-native, Gen AI* April 2024 – Dec 2025

- Awarded **\$6,000** at a startup competition for creating an app using Expo, React, and React-native as the frontend
- Implemented Tensorflow, specifically ICR, receipt, and food detection models

Real-time Edge Derivative Instance Segmentation | *Python, OpenCV, PyTorch, YOLOv8* May 2025

- Built a more accurate real-time instance segmentation pipeline using edge derivatives, YOLOv8, and COCO labels

Other Projects | *Web Development, Chrome Extensions, Mathematical Modeling* 2024 – 2025

Root2Success	SMBJ ML Pipeline (UWash)	Math Modeling Analysis
APUSH Website	Merck Analysis & MVP	LockedIn Extension
FRC Robotics '25	FTC Robotics '23	FTC Robotics '24

TECHNICAL SKILLS

Programming Languages: Java, JavaScript, HTML/CSS, TypeScript, Python, LaTeX
Frameworks/Libraries: Pandas, React, TensorFlow, OpenCV, Tailwind, Matplotlib, GSAP, Spring Boot
Skills: Web/App Dev, Data Science, Machine Learning, Computer Vision, Analysis

AWARDS

Chess Awards 2018 - Present

Earned: **>\$4,700** | Top 30 UltraBullet on Lichess | 4th @ World Open | 5th @ High School Nationals | 1st @ Liberty Bell | 4th @ National Chess Congress

3rd Place- Penn Mathematics Contest April 2025

1st Place- AMC 12B 2024 & 2025 | Certificate of Distinction and AIME Qualifier Nov 2024

Congressional App Challenge Finalist Nov 2024